Chapter-4

Agile Development And SCRUM

Agile Development and Scrum

- Agile emphasizes building working software that people can get hands on with quickly, versus spending a lot of time writing specifications up front.
- Agile focuses on small, cross-functional teams empowered to make decisions, versus big hierarchies and compartmentalization by function,.
- Agile focuses on rapid iteration, with as much customer input along the way as possible. .

fastest-growing Agile method

- One of the fastest-growing Agile methods is Scrum.
- It was formalized over a decade ago by Ken Schwaber and Dr. Jeff Sutherland, and it's now being used by companies large and small,
- including Yahoo!,
- Microsoft,
- Google,
- Lockheed Martin,
- Motorola,
- SAP, Cisco, GE Medical, CapitalOne and the US Federal Reserve.

SCRUM MODEL

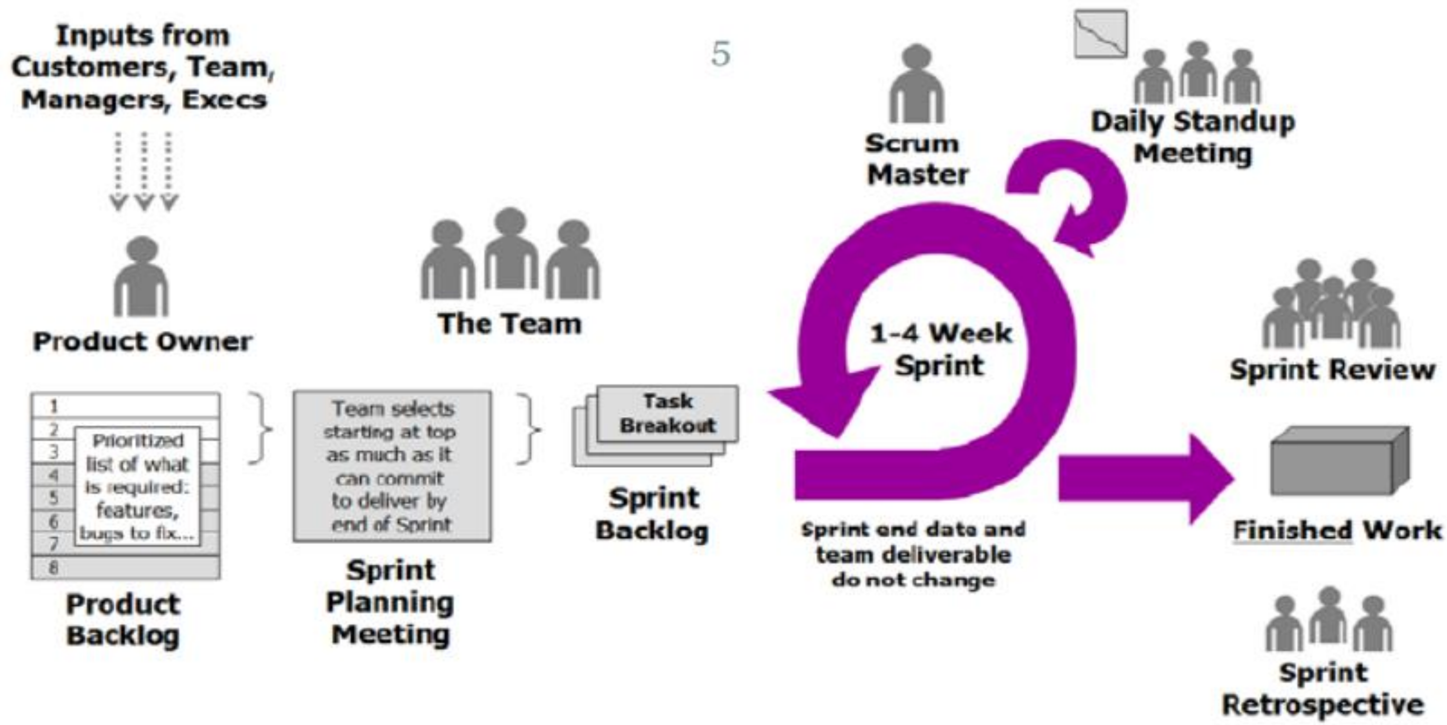

Figure 1. Scrum

SCRUM

- Scrum is an iterative, incremental process. Scrum structures product development in cycles of work called Sprints, iterations of work which are typically 1-4 weeks in length.
- The Sprints are of fixed duration – they end on a specific date whether the work has been completed or not, and are never extended.
- At the beginning of each Sprint, a cross-functional team selects items from a prioritized list of requirements, and commits to complete them by the end of the Sprint.
- Each work day, the team gathers briefly to report to each other on progress, and update simple visual representations of work remaining.

- At the end of the Sprint, the team demonstrates what they have built, and gets feedback which can then be acted upon in the next Sprint.
- Scrum emphasizes producing working product that at the end of the Sprint is really “done”; in the case of software, this means code that is fully tested and potentially shippable.

Scrum Roles

- In Scrum, there are three primary roles:
- 1. The Product Owner
- 2. Team Members
- 3. The ScrumMaster

Product Owner

- The Product Owner is responsible for taking all the inputs into what the product should be – from the customer or end-user of the product, as well as from Team Members and stakeholders – and translating them into a product vision.
- In some cases, the Product Owner and the customer are one and the same.
- Product owner is responsible for product backlog, manages release plan, adjusts priorities based on ROI and business value, accepts or rejects work results.

Team Members

- Team Members build the product that the customer is going to consume: the software, the website, or whatever it may be.
- The team in Scrum is typically five to ten people, although teams as large as 15 and as small as 3 commonly report benefits.
- The team should include all the expertise necessary to deliver the finished work – so, for example, the team for a software project might include roles (programmers, interface designers, testers, marketers, and researchers).

Scrum Master

- The Scrum Master is tasked with doing whatever is necessary to help the team be successful.
- The Scrum Master is not the manager of the team; he or she serves the team, by helping remove blocks to the team's success, facilitating meetings, and supporting the practice of Scrum.
- Some teams will have someone dedicated fully to the role of Scrum Master, while others will have a team member play this role (carrying a lighter load of regular work when they do so).

- Great Scrum Masters have come from all backgrounds and disciplines: Project Management, Engineering, Design, Testing.
- The Scrum Master and the Product Owner probably shouldn’t be the same individual; at times, the Scrum Master may be called upon to push back on the Product Owner (for example, if they try to introduce new requirements in the middle of a Sprint).

Product Backlog

- The first step in Scrum is for the Product Owner to articulate the product vision.
- This takes the form of a prioritized list of what's required, ranked in order of value to the customer and business, with the highest value items at the top of the list. This is called the Product Backlog, and it exists (and evolves) over the lifetime of the product (figure 2).

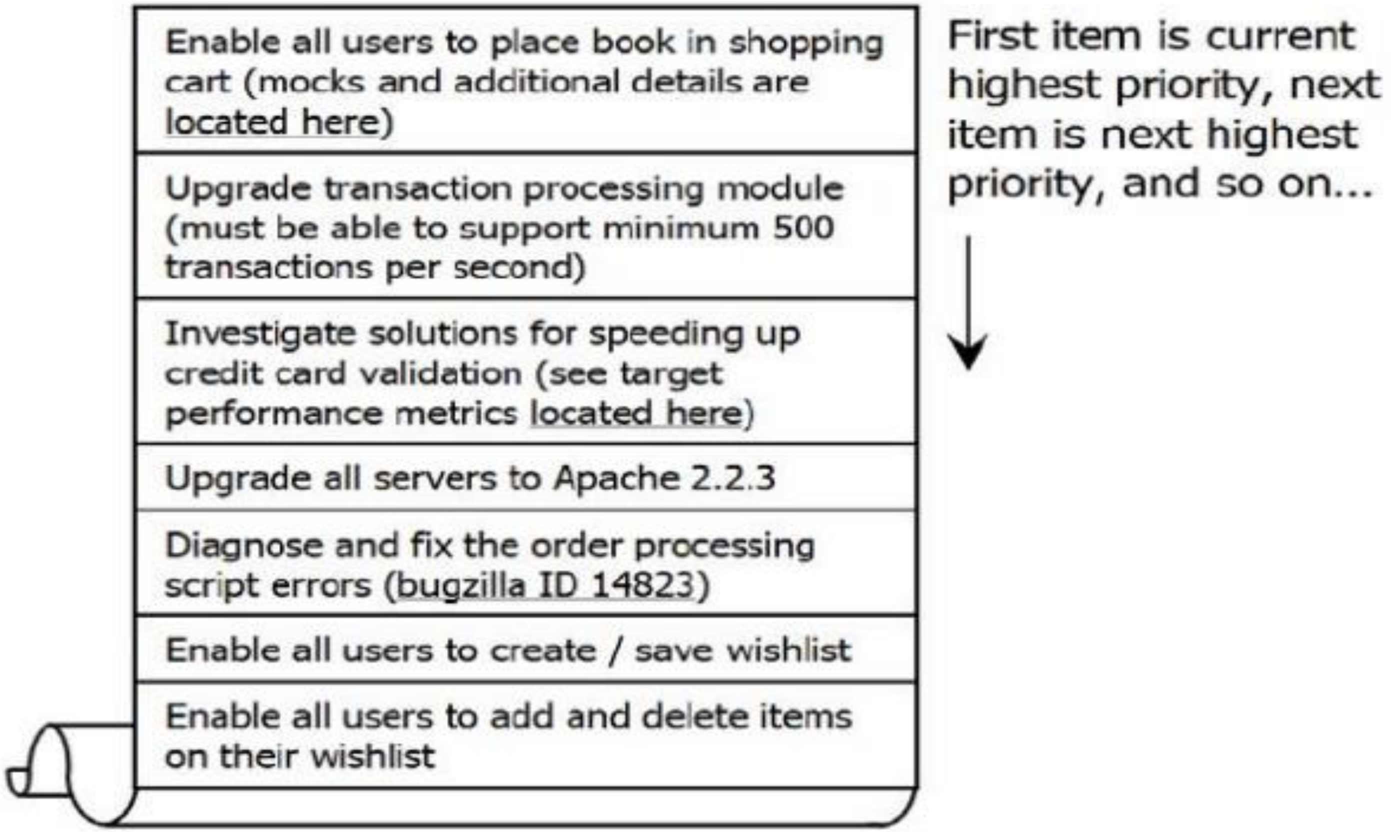

Figure 2. The Product Backlog

Sprint Planning Meeting

- In the first part of the Sprint Planning Meeting, the Product Owner and Scrum Team (with facilitation from the Scrum Master) review the Product Backlog, discussing the goals and context for the items on the Backlog, and providing the Scrum Team with insight into the Product Owner's thinking.
- In the second part of the meeting, the Scrum Team selects the items from the Product Backlog to commit to complete by the end of the Sprint, starting at the top of the Product Backlog (in others words, starting with the items that are the highest priority for the Product Owner) and working down the list in order.

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- This is one of the key practices in Scrum: the team decides how much work they will commit to complete, rather than having it assigned to them by the Product Owner. This makes for a much more reliable commitment
- The Sprint Planning meeting will often last a number of hours
- the team is making a very serious commitment to complete the work, and this commitment requires careful thought to be successful.

- The team will begin by estimating how much time each member has for Sprint-related work – in other words, their average workday minus the time they spend doing things like critical bug-fixes and other maintenance, attending meetings, doing email, taking lunch breaks, and so on. For most people this works out to 4-6 hours of time per day available for Sprint-related work. (Figure 3.)

Sprint Length	2 weeks
Workdays During Sprint	10 days

Team Member	**Available Days During Sprint***	**Available Hours Per Day**	**Total Available Hours During Sprint**
Tracy	9 days	4 hours	36 hours [=9 x 4]
Sanjay	10 days	5 hours	50 hours
Phillip	10 days	4 hours	40 hours
Jing	8 days	5 hours	40 hours

*Net of vacation, other days out of the office

Figure 3. Estimating Available Hours

Sprint backlog

- Once the time available is determined, the team starts with the first item on the Product Backlog
- In other words, the Product Owner’s highest priority item – and working together, breaks it down into individual tasks, which are recorded in a document called the Sprint Backlog (figure 4).
- At the end of the meeting, the team will have produced a list of all the tasks, and for each task who has signed up to complete it and how much time they estimate it will take (typically in hours or fractions of a day).

Backlog Item	Task	Owner	Initial Time Estimate
Upgrade transaction processing module	Configure database and space IDs for Trac	Sanjay	4 hours
	Use test data to tune the learning and action model	Jing	2 hours
	Setup a cart server code to run as apache server	Philip	3 hours
	Implement pre-Login Handler	Tracy	3 hours
Investigate solutions for speeding up credit card validation	Merge DCP code and complete layer-level tests	Jing	5 hours
	Complete machine order for pRank	Jing	4 hours
	Change DCP and reader to use pRank http API	Tracy	3 hours

Figure 4. Sprint Backlog

Scrum Change

- One of the key pillars of Scrum is that once the Scrum Team makes its commitment, the Product Owner cannot add new requests during the course of the Sprint. This means that even if halfway through the Sprint the Product Owner decides that they want to add something new, he or she cannot make changes until the start of the next Sprint.
- If an external circumstance appears that significantly changes priorities, and means the team would be wasting its time if it continued working, the Product Owner can terminate the Sprint; this means the team stops all the work they are doing, and starts over with a Sprint Planning meeting, and so forth.

Daily Standup Meeting

- Once the Sprint has started, the Scrum Team engages in another of the key Scrum practices: The Daily Stand-Up Meeting.
- This is a short (15 minute) meeting that happens every workday at an appointed time, and everyone on the Scrum Team attends; in order to ensure it stays brief, everyone stands (hence “Stand-Up Meeting”).
- It’s the team’s opportunity to report to itself on progress and obstacles.

One by one, each member of the team reports just three things to the other members of the team:

1. What they were able to get done since the last meeting.
2. what they're aiming to get done by the next meeting.
3. any blocks or obstacles that are in their way.

There's no discussion during the Daily Stand-Up Meeting, just the reporting of the three key pieces of information; if discussion is required, it takes place right after the meeting

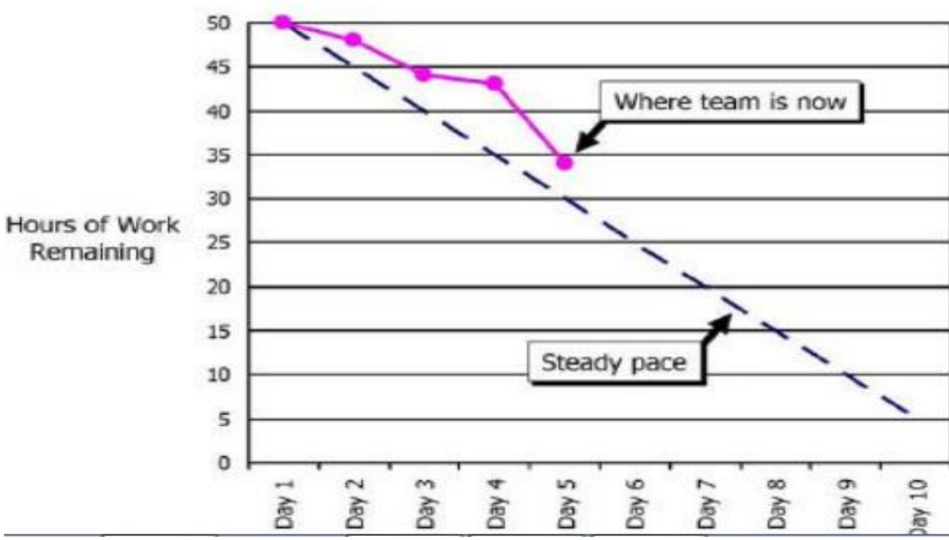
Role of participants during meeting

- The Scrum Master makes note of the blocks, and then helps team members to resolve them after the meeting.
- The Product Owner, Managers, and other stakeholders can attend the meeting, but they should refrain from asking questions or opening discussion until after the meeting concludes
- everyone should be clear that the team is reporting to each other, not to the Product Owner, Managers or Scrum Master. While it’s non-standard, some teams find it useful to have the Product Owners join and give a brief daily report of their own activities to the team.

After meeting

- After the meeting, the team members update the amount of time remaining to complete each of the tasks that they’ve signed up for on the Sprint Backlog.
- The important thing is that it show the team their actual progress towards their goal – and not in terms of how much time has been spent so far (an irrelevant fact, as far as Scrum is concerned), but in terms of how much work remains – what separates the team from their goal.
- If the curve is not tracking towards completion at the end of the Sprint, then the team needs to either pick up the pace, or simplify and cut down what it’s doing.

		Hours of Work Remaining on Each Day of the Sprint									
Task	Task Owner	Day 1	Day 2	Day 3	Day 4	Day 5	Day 6	Day 7	Day 8	Day 9	Day 10
Configure database and space IDs for Trac	Sanjay	4	4	3	1	0					
Use test data to tune the learning and action model	Jing	2	2	2	2	1					
Setup a cart server code to run as apache server	Phillip	3	3	5	2	0					
Implement pre-Login Handler	Tracy	3	3	3	3	3					
Merge DCP code and complete layer-level tests	Jing	5	5	2	2	2					
Complete machine order for pRank	Jing	4	4	3	3	3					
Change DCP and reader to use pRank http API	Tracy	3	3	0	0	0					
	Total	50	48	44	43	34					



This information is recorded on a graph called the Sprint Burndown Chart (figure 5). It shows, each day, how much work (measured in hours or days) remains until the team's commitment is completed. Ideally, this should be a downward sloping graph that is on a trajectory to hit zero on the last day of the Sprint. And while sometimes it looks like that, often it doesn't.

Sprint status

- One of the core tenets of Scrum is that the duration of the Sprint is never extended – it ends on the assigned date regardless of whether the team has completed the work it committed to or not.
- If the team has not completed their Sprint Goal, they have to stand up at the end of the Sprint and acknowledge that they did not meet their commitment. The idea is that this creates a very visible feedback loop, and teams are forced to get better at estimating what they are capable of accomplishing in a given Sprint, and then delivering it without fail.

- Teams will typically over-commit in their first few Sprints and fail to meet their Sprint Goal; they might then overcompensate and under commit, and finish early; but by the third or fourth Sprint, teams will typically have figured out what they're capable of delivering, and they'll meet their Sprint goals reliably after that.
- Teams are encouraged to pick one duration for their Sprints (say, 2 weeks) and not change it frequently – a consistent duration helps the team learn how much it can accomplish, and it also helps the team achieve a rhythm for their work (this is often referred to as the "heartbeat" of the team).

Sprint Review

- After the Sprint ends, there is the Sprint Review, where the team demos what they've built during the Sprint.
- Present at this meeting are the Product Owner, Team Members, and ScrumMaster, plus customers, stakeholders, experts, executives, and anyone else interested. This is not a "presentation" the team gives – there are no PowerPoints, and typically no more than 30 minutes is spent preparing for it – it's literally just a demo of what's been built, and anyone present is free to ask questions and give input. It can last 10 minutes, or it can last two hours – whatever it takes to show what's been built and get feedback

Sprint Retrospective

- Following the Sprint Review, the team gets together for the Sprint Retrospective.
- It's an opportunity for the team to discuss what's working and what's not working, and agree on changes to try. The Scrum Team, the Product Owner, and the ScrumMaster will all attend, and a neutral outsider will facilitate the meeting; a good approach is for Scrum Masters to facilitate each others' retrospectives, which enables crosspollination among teams. This is a practice that some teams skip, and that's unfortunate because it's one of the most important tools for making Scrum successful

Starting the Next Sprint

- Following the Sprint Review Meeting, the Product Owner takes all the input, as well as all new priorities that have appeared during the Sprint, and incorporates them into the Product Backlog; new items are added, and existing ones are modified, reordered, or deleted.
- Once this updating of the Product Backlog is complete, the cycle is ready to begin all over again, with the next Sprint Planning Meeting.
- One practice many teams find useful is to hold a Prioritization Meeting toward the end of each Sprint, to review the Product Backlog for the upcoming Sprint with the Product Owner.

- In addition to giving the team an opportunity to suggest items the Product Owner may not be aware of – technical maintenance, for example – this meeting also kicks off any preliminary thinking that’s required before the Sprint Planning Meeting.
- There’s no downtime between Sprints – teams will often go from a Sprint Review one afternoon into the next Sprint Planning Meeting the following morning.
- One of the values of Agile development is “sustainable pace”, and only by working regular hours at a reasonable level of intensity can teams continue this cycle indefinitely